Auctions with Risk Reserves for Scarce AI Research

Zhenning Wang · Qianshuo (Aaron) Wang
Duke Kunshan University • Professor Luyao Zhang • Team FP08 • Final Project, October 2026

昆山杜克大学 DUKE KUNSHAN UNIVERSITY

KEY FINDING **Hybrid is conditionally, not universally, preferred.**
Added risk coverage must justify displaced private research value.

01 QUESTION, MODEL & MERITS

When should an institution reserve a slot?

Can a verified risk reserve improve scarce AI research allocation relative to a pure first-price auction?

Application: an asset manager during earnings season. The 20-PM / 4-slot setting is a research assumption, not documented T. Rowe Price practice.

20 portfolio managers, 4 identical slots

One slot per PM. Private value $v_i \in [0,100]$.
Public risk score $r_i = 100E_iS_i$ is frozen before bids.

PURE FPA Four highest bids win and pay their own bids.

HYBRID Highest $r_i \geq 60$ gets one free reserve slot.
Auction the other three. No eligible PM: auction all four.

FCFS Four earliest requests win for free, by exogenous arrival.

$W = V + \lambda C$ Hybrid beats Pure FPA iff $\lambda\Delta C > L$
V = total allocated private value C = total allocated risk
$\Delta C = C(H) - C(P)$ $L = V(P) - V(H)$
Payments are internal transfers, reported separately.

Intellectual Merits: inherited foundations

ATHEY, COEY & LEVIN [1]
Auction set-asides and subsidies

SIVAN ET AL. [2]
Dynamic prices for scarce ML quotas

KARMA [3]
History-based credits for repeated demand

FP08: one current-risk reserve plus three first-price slots
A one-shot comparison with fixed bids and explicit displacement cost

Supporting evidence: 1,000 matched synthetic batches
Conditional welfare test $\lambda\Delta C > L$, with strategic and score sensitivity

Bounded contribution: a transparent application-level comparison.
No new equilibrium, field effectiveness, or avoided-loss result.

02 METHOD & EVIDENCE

Matched simulation, fixed assumptions

1,000 synthetic batches; seed 20260926. The same requests, values and risk scores enter all three rules.
Stylized multi-slot bids: $b_i = 0.8v_i$ in both auctions.

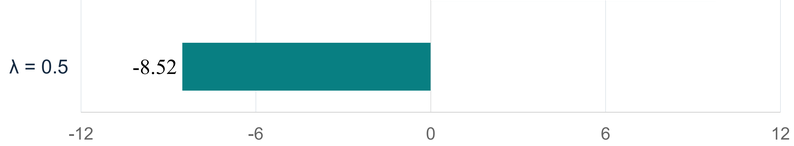
Assumed bidding rule, not a multi-slot equilibrium.

+9.78
mean ΔW at $\lambda = 1$
Monte Carlo SE = 0.89

L = 26.82
Displaced private value

ΔC = +36.59
Additional risk coverage

Mean Hybrid − Pure FPA welfare



43.8% of batches strictly improve. $\lambda^* \approx 0.733$

λ^* is the ratio of mean loss to mean coverage gain.
It is model-implied, not an estimated firm preference.

Two completed sensitivity checks

1 BID SHADING: PM11, SAME REQUEST BATCH

0.8v → 0.9v changes the Hybrid allocation.
ΔW: −15.85 → +4.50
Sensitivity near an allocation margin; not equilibrium evidence.

2 SCORE THRESHOLD: BATCH 713, PM20

Observed score 59.9 → 60.0 crosses $r^* = 60$,
activates the reserve and changes the winning set.

03 IMPACTS & FUTURE RESEARCH

Practical Impacts: intended SDG links

TARGET 9.5
Research capacity and innovation

TARGET 16.6
Accountable, transparent institutions

MECHANISM
Validate risk before bidding. One free risk reserve plus three auctioned slots.

STAKEHOLDERS
Portfolio managers · Risk teams · Computing operations · Clients

SIMULATED INDICATORS

Pure FPA		Hybrid	
C = 101.35	Eligible access = 20.29%	C = 137.95	Eligible access = 66.70%

POSSIBLE HARMS	PROPOSED SAFEGUARDS
Score gaming or classification error	Independent audits and r* tests
Biased access and data leakage	Decision logs, correction and appeals
Coverage mistaken for avoided loss	Shadow deployment before adoption

Allocation effects are simulated. SDG and financial benefits are intended, not demonstrated.

Future Research Roadmap

CURRENT LIMITATION	PLANNED STUDY	NEXT TEST
ECONOMICS Fixed 0.8v bidding	Endogenous multi-slot strategic bidding	Does the welfare ranking survive?
COMPUTER SCIENCE Unvalidated risk score	Score validation, audit and correction	Robustness around r* = 60
BEHAVIORAL SCIENCE No collected human bids	Gain/loss framing experiment	Does framing change shading?

BEHAVIORAL EVIDENCE: PLANNED / EXPLORATORY

Stronger evidence for conditional institutional adoption

TAKEAWAY **A reserve is justified only when the institution values added risk coverage enough to offset displaced private value.**

ACKNOWLEDGMENTS & CONTRIBUTIONS

Prof. **Luyao Zhang**; Prof. **Kenneth S. (Ken) Rogerson; Han Zhang; Temur Akhtamjonov; Mustafa Ayub Khan; Gihun Lee; Longfei Jing; Jiayang Sun**; and the COMSCI/ECON 206 community. Field-learning settings: Tencent's Shanghai office and Shanghai Science and Technology Museum (no project data or endorsement).
Zhenning Wang: behavioral interface and impact analysis. **Qianshuo (Aaron) Wang**: model and reproducible computation. AI assistance disclosed in the paper; authors retain responsibility.

THREE CLOSEST PAPERS

[1] Athey, Coey & Levin (2013). Set-Asides and Subsidies in Auctions. AEJ: Microeconomics 5(1), 1–27. doi:10.1257/mic.5.1.1
[2] Sivan et al. (2026). Quota Marketplace: Dynamic Pricing for Efficient Allocation of ML Training Resources. OSDI, 1223–1241. usenix.org/conference/osdi26/presentation/sivan
[3] Vuppalapati et al. (2023). Karma: Resource Allocation for Dynamic Demands. OSDI, 645–662. usenix.org/conference/osdi23/presentation/vuppalapati

CURRENT ARTIFACTS

GitHub: code & reproducibility
github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning
Colab: runnable notebook
colab.research.google.com/drive/1I-J52wxGaAIwJK_LjtghBnXS2GxY0U2V
Hugging Face: planned behavioral interface
huggingface.co/spaces/dku-comsci-econ206-2026/Final_Zhenning_Qianshuo_PM_Bid

SDG icons: United Nations, https://www.un.org/sustainabledevelopment. The content of this publication has not been approved by the United Nations and does not reflect the views of the United Nations or its officials or Member States.